Sheet 10

A1.1

$$H_{Nose} = \sum_{i=1}^{N} \frac{1}{2m_i} \vec{p}_i^{\,2} + U(\{\vec{r}_i\}) + \frac{p_s^2}{2Q} + \frac{3N+1}{\beta} \ln(s)$$

$$\vec{p}_{Nose} = \sum_{i=1}^{N} s\vec{p}_i \Rightarrow \vec{p}_{N,i} = s\vec{p}_i \quad , \quad \sum_{i=1}^{N} \vec{r}_i = \vec{r}_N$$

$$H_{Nose} = \sum_{i=1}^{N} \frac{1}{2m_i s^2} p_{N,i}^2 + U(\vec{r}_N) + \frac{p_s^2}{2Q} + \frac{3N+1}{\beta} \ln(s)$$

A1.2

$$Z^{micro}_{Nose} = \int d^{3N}\vec{p}_N \, d^{3N}\vec{r}_N \, ds \, dp_s \, \delta(H_{nose} - E)$$

$$= \int d^{3N}\vec{p}_N \, d^{3N}\vec{r}_N \, ds \, dp_s \, \delta\left(\sum_{i=1}^{N} \frac{1}{2m_i s^2} p_{N,i}^2 + U(\vec{r}_N) + \frac{p_s^2}{2Q} + \frac{3N+1}{\beta} \ln(s) - E\right)$$

$d\vec{p}_N = s\,d\vec{p}$

$$= \int d^{3N}\vec{p} \, d^{3N}\vec{r} \, ds \, dp_s \, \delta\left(\sum_{i=1}^{N} \frac{1}{2m_i} p_i^2 + U(\vec{r}) + \frac{p_s^2}{2Q} + \frac{3N+1}{\beta} \ln(s) - E\right) \cdot s^{3N}$$

$$\rightarrow \delta(f(x)) = \frac{\delta(x - x_0)}{|f'(x_0)|} \Rightarrow \delta(H(s) - E) = \frac{\delta\left(s - \underbrace{\exp\left[\beta\frac{\left(E - \left(\sum \frac{1}{2m_i s^2} p_{N,i}^2 + U(\vec{r}_N)\right) - \frac{p_s^2}{2Q}\right)}{3N+1}\right]}_{= S_0}\right)}{\frac{1}{S_0} \cdot \frac{3N+1}{\beta}} = \frac{\delta(s - S_0) \cdot S_0}{3N+1} \beta$$

$$\rightarrow \int d^{3N}\vec{p} \, d^{3N}\vec{r} \, dp_s \, ds \, s^{3N} \, \delta(H(s) - E)$$

$$= \int d^{3N}\vec{p} \, d^{3N}\vec{r} \, dp_s \, S_0^{3N+1} \cdot \frac{\beta}{3N+1} = \int d^{3N}p \, d^{3N}r \, dp_s \, \frac{\beta}{3N+1} \exp\left[\frac{\beta}{3N+1}\left(E - H_{real} - \frac{p_s^2}{2Q}\right)\right]^{3N+1}$$

$$= \int d^{3N}p \, d^{3N}r \, dp_s \, \frac{\beta}{3N+1} \exp\left[\beta\left(E - H_{real} - \frac{p_s^2}{2Q}\right)\right]$$

$$= \int d^{3N}p \, d^{3N}r \, e^{-\beta H_{real}} \underbrace{\int dp_s \, \frac{\beta}{3N+1} \, e^{\beta\left(E - \frac{p_s^2}{2Q}\right)}}_{C}$$

$$= C \, Z^{can}_{real}$$

expectation value:

$$\langle O \rangle = \frac{1}{Z^{can}_{real}} \int O(r, p) \, d^{3N}p \, d^{3N}r \, e^{-\beta H_{real}} =$$

$$= \frac{1}{Z^{can}_{real}} \int O(r_N, p_N) \, d^{3N}p_N \, d^{3N}r_N \, ds \, dp_s \, \delta(H - E) \cdot C$$

$$= \frac{1}{Z^{micro}_{Nose}} \int O(r_N, p_N) \, d^{3N}p_N \, d^{3N}r_N \, ds \, dp_s \, \delta(H - E)$$